CLOUD COMPUTING

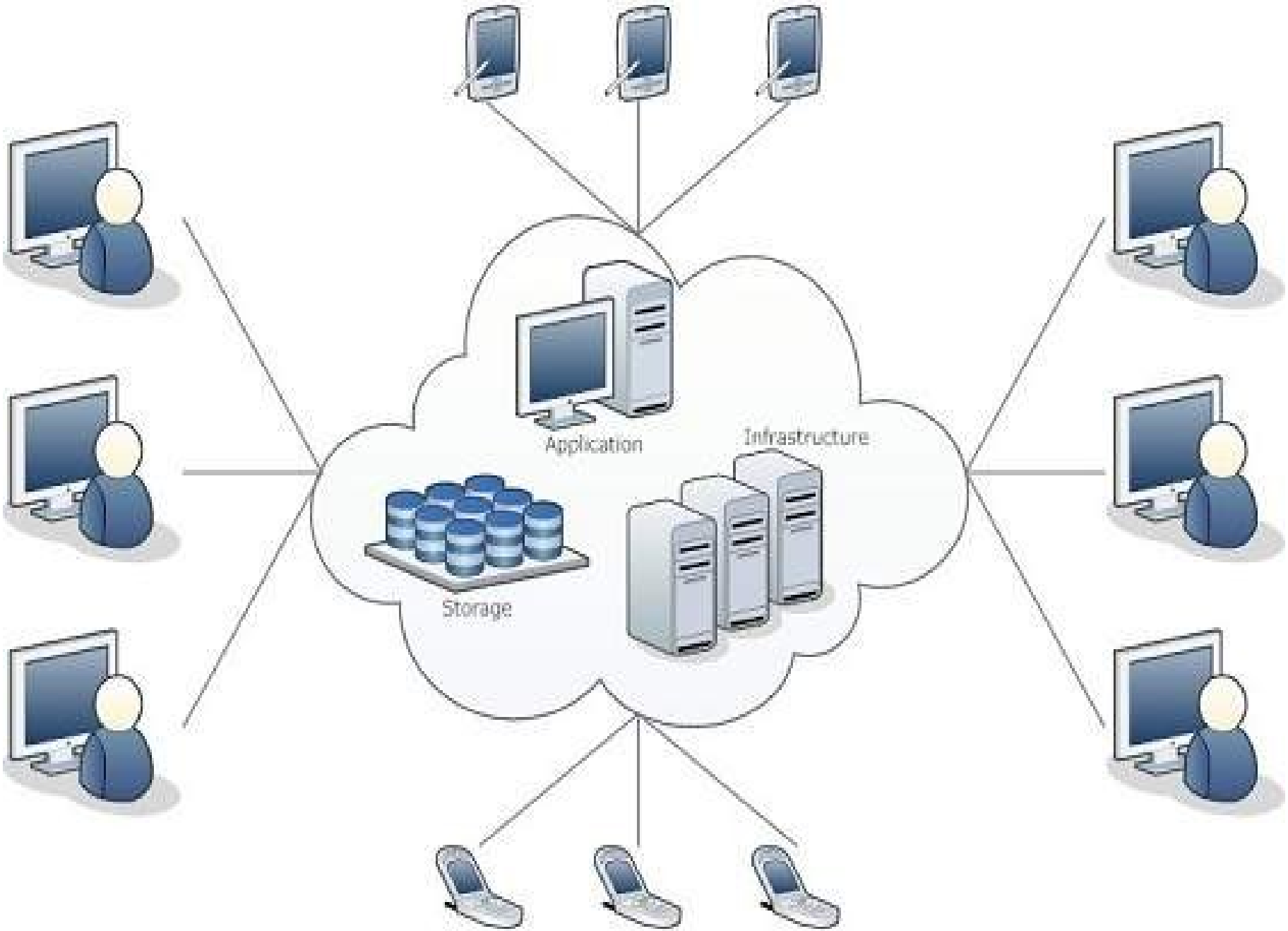
UNIT NO. 1

What Is Cloud Computing?

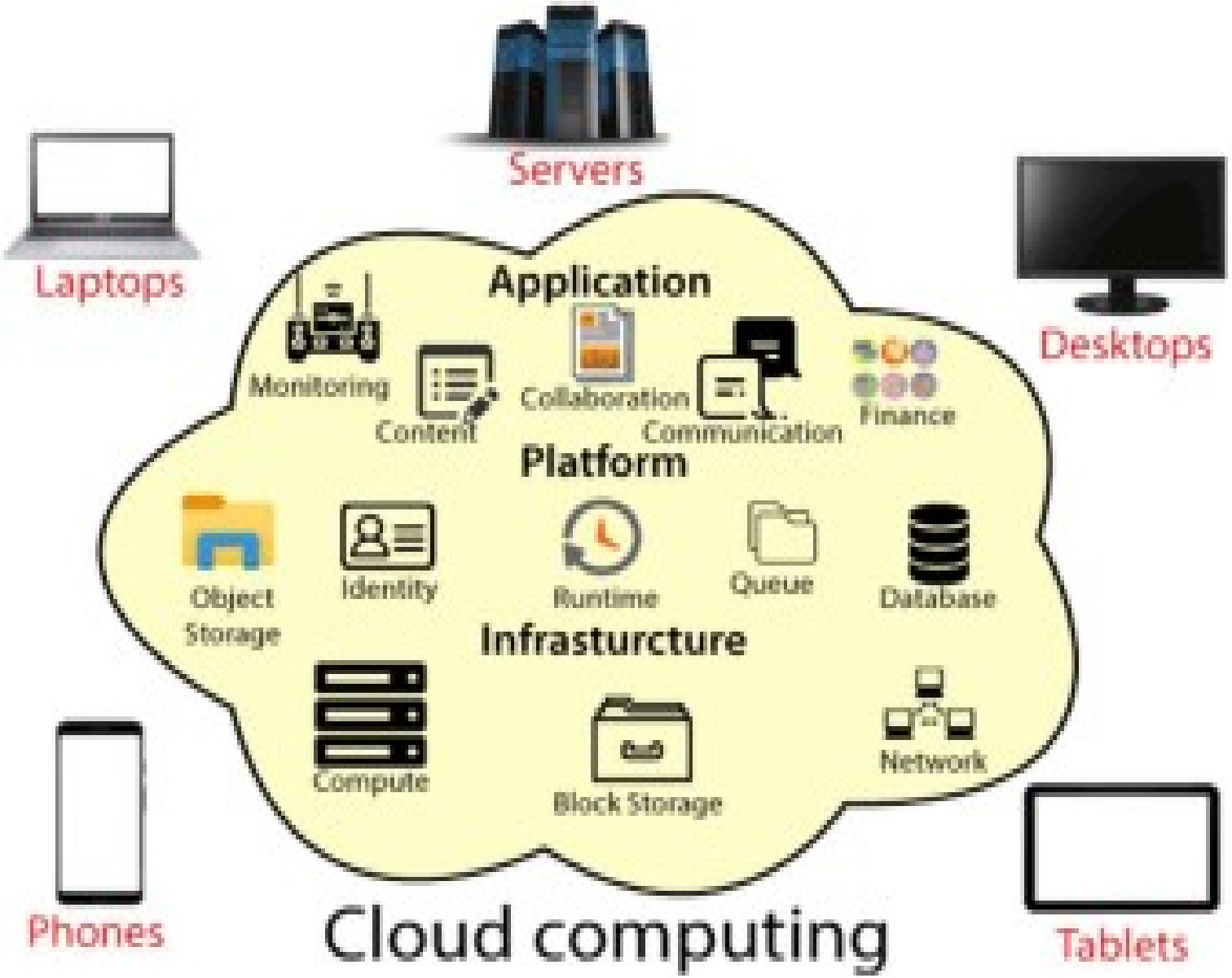
- Cloud computing is a model for enabling *convenient*, *on-demand network access* to a *shared pool* of *configurable computing resources* (e.g., networks, servers, storage, applications, and services)

- Definition 2 :Cloud computing is the delivery of different services through the Internet. These resources include tools and applications like data storage, servers, databases, networking, and software.
- Rather than keeping files on a proprietary hard drive or local storage device, cloud-based storage makes it possible to save them to a remote database.
- As long as an electronic device has access to the web, it has access to the data and the software programs to run it.

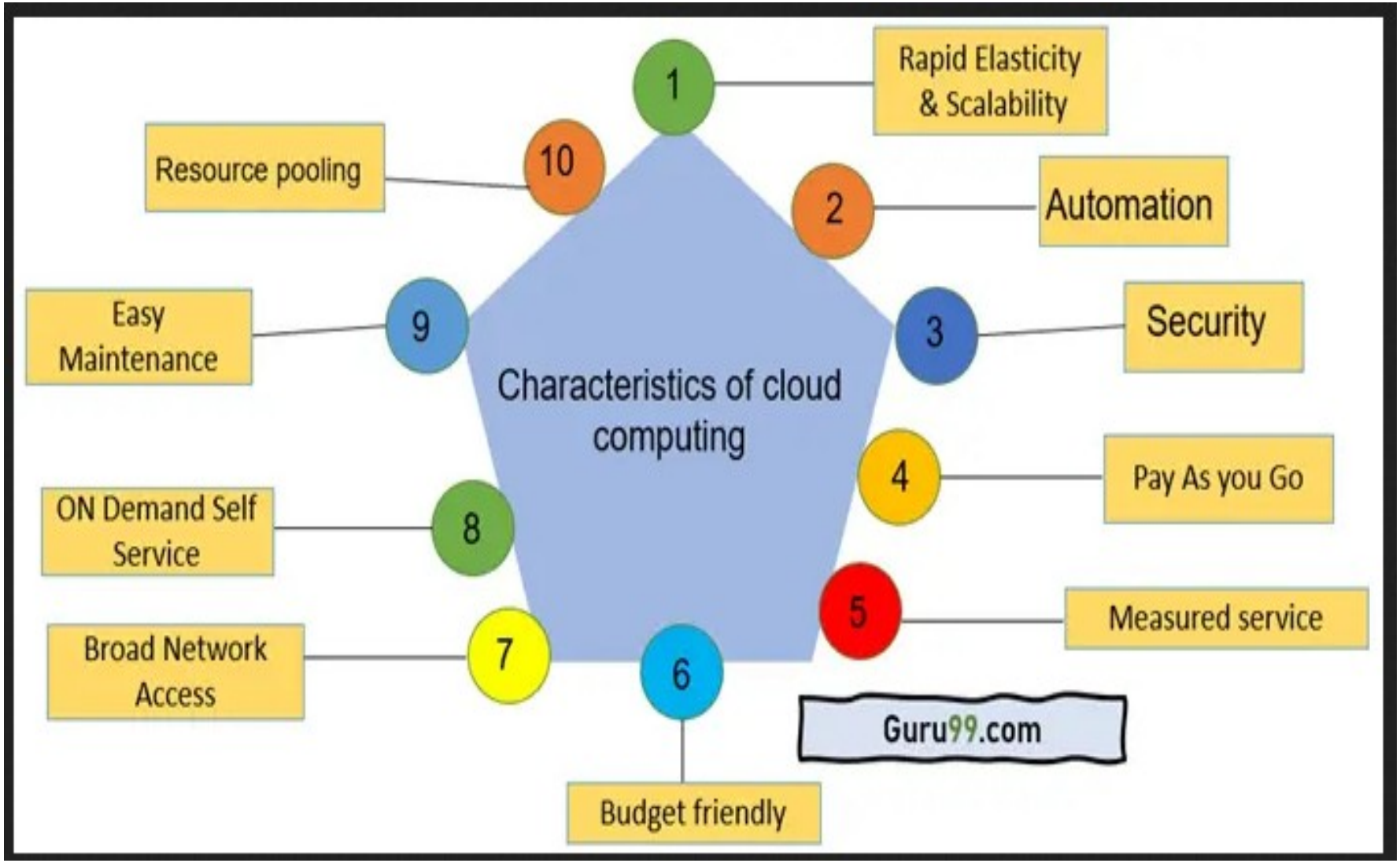
- Since cloud computing began, the world has witnessed an explosion of cloud-based applications and services in IT, which continue to expand.
- In 2021, almost every small and large application we use resides on the cloud, helping us save storage space, expenses, and time.


Application
Infrastructure
Storage

- However, if we choose Cloud Computing, a cloud vendor is responsible for the hardware purchase and maintenance.
- They also provide a wide variety of software and platform as a service.
- We can take any required services on rent.
- The cloud computing services are charged based on usage.


Servers
Laptops
Desktops
Application
Monitoring
Content
Collaboration
Communication
Finance
Platform
Object Storage
Identity
Runtime
Queue
Database
Infrasturcture
Compute
Block Storage
Network
Phones
Tablets
Cloud computing

CHARACTERISTICS OF CLOUD COMPUTING



- **On-demand self-services:** The Cloud computing services does not require any human administrators, user themselves are able to provision, monitor and manage computing resources as needed.
- **Broad network access:** The Computing services are generally provided over standard networks and heterogeneous devices.
- **Rapid elasticity:** The Computing services should have IT resources that are able to scale out and in quickly and on as needed basis. Whenever the user require services it is provided to him and it is scale out as soon as its requirement gets over.

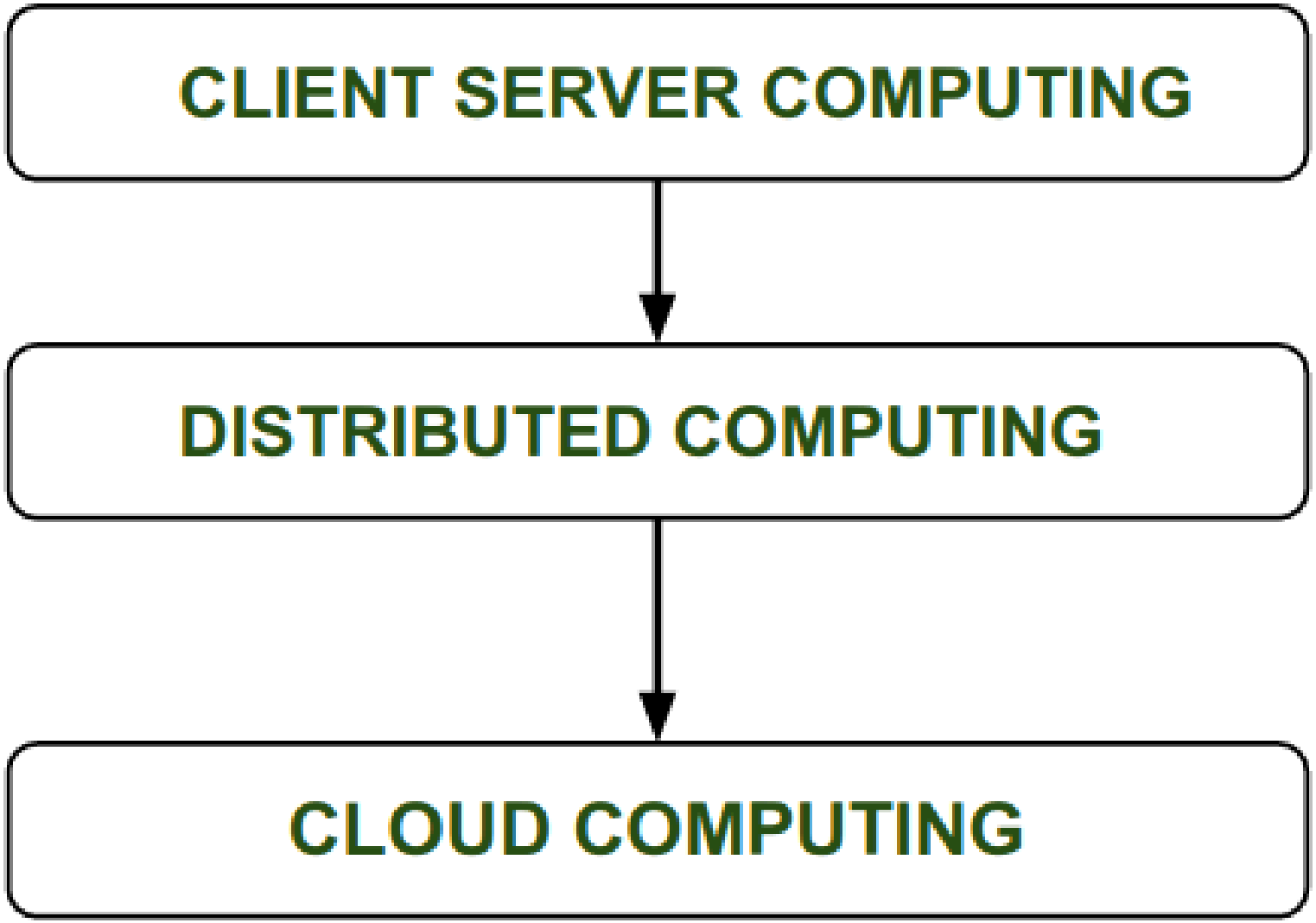
- **Resource pooling:** The IT resource (e.g., networks, servers, storage, applications, and services) present are shared across multiple applications and occupant in an uncommitted manner. Multiple clients are provided service from a same physical resource.
- **Measured service:** The resource utilization is tracked for each application and occupant, it will provide both the user and the resource provider with an account of what has been used. This is done for various reasons like monitoring billing and effective use of resource.
- **Multi-tenancy:** Cloud computing providers can support multiple tenants (users or organizations) on a single set of shared resources.

- **Virtualization:** Cloud computing providers use virtualization technology to abstract underlying hardware resources and present them as logical resources to users.
- **Flexible pricing models:** Cloud providers offer a variety of pricing models, including pay-per-use, subscription-based, and spot pricing, allowing users to choose the option that best suits their needs.

- **Security:** Cloud providers invest heavily in security measures to protect their users' data and ensure the privacy of sensitive information.
- **Automation:** Cloud computing services are often highly automated, allowing users to deploy and manage resources with minimal manual intervention.

History of Cloud Computing

- Before Cloud Computing was come into existence, client Server Architecture was used where all the data and control of client resides in Server side.
- If a single user want to access some data, firstly user need to connect to the server and after that user will get appropriate access. But it has many disadvantages.


CLIENT SERVER COMPUTING
DISTRIBUTED COMPUTING
CLOUD COMPUTING

- So, After Client Server computing, **Distributed Computing was come into existence, in this type of computing all computers are networked together with the help of this, user can share their resources when needed.**
- It also has certain limitations.
- So in order to remove limitations faced in distributed system, cloud computing was emerged.

Early Foundations (1960s–1980s)

- In the **1960s, computer scientist John McCarthy** proposed the idea that computing could someday be organized as a public utility.
- During the **1970s, time-sharing systems** allowed multiple users to share computing resources, forming the foundation of remote computing services.
- In the **1980s, the growth of the internet and client-server architecture** enabled remote access to data and applications.

Virtualization and Grid Computing (1990s)

- In the **1990s, virtualization technology** allowed multiple operating systems to run on a single physical machine, improving resource utilization.
- **Grid computing emerged, enabling distributed computing across multiple systems** — a precursor to modern cloud infrastructure.

Birth of Modern Cloud Computing (2000s)

- In 2002, Amazon Web Services (AWS) launched web-based storage and computing services.
- In 2006, Amazon introduced Elastic Compute Cloud (EC2), marking the beginning of commercial cloud computing.
- Google and Microsoft soon entered the market with Google Cloud and Microsoft Azure.

Rapid Growth and Adoption (2010s)

- Cloud services expanded into:
- IaaS (Infrastructure as a Service)
- PaaS (Platform as a Service)
- SaaS (Software as a Service)
- Businesses adopted cloud for scalability, cost reduction, and global access.

Present Era (2020s–Today)

- Cloud computing now integrates:
- Artificial Intelligence and Machine Learning
- Big Data Analytics
- Internet of Things (IoT)
- Edge and Hybrid Cloud
- Major providers include AWS, Microsoft Azure, Google Cloud, IBM Cloud, and Oracle Cloud.

Cloud Computing Challenges

- **1. Data Security and Privacy**
- Data security is a major concern when switching to cloud computing. User or organizational data stored in the cloud is critical and private.
- Storing data on remote cloud servers raises concerns about unauthorized access, data breaches, and privacy protection.

Reliability and Downtime

- Cloud services depend completely on internet and service providers.
- If the provider’s servers or network fail, users may temporarily lose access to applications and data.

Vendor Lock-in

Once an organization adopts a specific cloud provider, moving to another provider becomes difficult due to differences in platforms, tools, and data formats.

Limited Control over Infrastructure

- In cloud computing, users do not have physical control over servers and storage, which makes it hard to customize hardware configurations.

Data Transfer and Bandwidth Cost

- Uploading and downloading large amounts of data requires high bandwidth, and cloud providers often charge extra for data transfer.

- **6. High Dependence on Network**
- Since cloud computing deals with provisioning resources in real-time, it deals with enormous amounts of data transfer to and from the servers.
- This is only made possible due to the availability of the high-speed network.

- **7. Lack of Knowledge and Expertise**
- Due to the complex nature and the high demand for research working with the cloud often ends up being a highly tedious task. It requires immense knowledge and wide expertise on the subject.
- Although there are a lot of professionals in the field they need to constantly update themselves. Cloud computing is a highly paid job due to the extensive gap between demand and supply.

VISION OF CC

- The vision of cloud computing extends far beyond the simple concept of remote data storage; it encompasses a revolutionary approach to managing, accessing, and utilizing IT resources.

- **Scalability and Flexibility**: One of the core tenets of cloud computing is its unparalleled scalability.
- Cloud resources can be rapidly scaled up or down to meet fluctuating demands, reducing the need for hefty upfront investments in hardware and infrastructure.
- This flexibility allows organizations to adapt swiftly to changing market conditions and seasonal variations.

- **Cost-Efficiency:** Cloud computing is inherently cost-effective.
- It eliminates the need for significant capital expenditures on on-premises hardware.
- Users only pay for the resources they consume, making it an attractive option for businesses of all sizes.

- **Accessibility**: The cloud's ubiquity ensures that users can access their data, applications, and services from virtually anywhere with an internet connection.
- This promotes remote work, global collaboration, and enhanced customer reach. Regardless of geographical constraints, cloud computing enables seamless connectivity.

- **Reliability and Redundancy**: Cloud service providers invest heavily in redundancy and data backup, guaranteeing high levels of data integrity and minimizing downtime.
- This reliability ensures that critical operations can continue without disruption, instilling confidence in users and businesses.

- **Innovation and Agility**: Cloud computing empowers innovation by providing developers with the tools, platforms, and services needed to rapidly build and deploy applications.

- The agility it offers allows organizations to respond to market changes, customer demands, and competitive pressures with unprecedented speed and efficiency.

- **Security and Compliance**: Cloud providers prioritize security measures and compliance standards, often exceeding the capabilities of individual organizations.
- Users can leverage these robust security frameworks to enhance their own security postures and comply with industry-specific regulations, thereby reducing the burden of compliance.

- **Environmental Sustainability**: Cloud computing contributes to environmental sustainability through resource optimization.
- By sharing and efficiently managing resources within data centers, cloud providers often boast a smaller carbon footprint compared to traditional on-premises data centers.
- This aligns with global efforts to reduce the environmental impact of technology infrastructure.

DRIVING FACTORS TOWARDS CLOUD COMPUTING

1) Latest Software –

- Thanks to cloud based SaaS, customers can lay their hands on the fresh software no sooner than it is ready for release. One can instantly transfer new functionalities existing software and features by way of updates.

2) Minimize IT infrastructure –

- The most visible change of cloud is reduction of existing IT infrastructure without any impact on capabilities. Many organizations have been able to entirely eliminate the need for onsite infrastructure by reducing staff expenditure and maintenance costs of physical infrastructure.

- **3) Flexibility of costs –**
- Flexibility stands as one of the pivotal driving factors of cloud computing adoption. Unlike traditional systems, cloud solutions offer remarkable flexibility in payment methods.
- Users pay solely for the utilized infrastructure and server capabilities. Additional capacity can be readily provisioned or decommissioned, aligning with demand fluctuations—eliminating instances of idle, unutilized capacity.

4) Remarkable availability –

- The cloud ensures incessant availability of both data and applications, regardless of users' geographical location.

5) Enhanced mobility –

- Cloud based applications facilitate employees to work from anywhere with help of internet enabled smart devices. Employees can even work while on the go.

- **6) Enhanced collaboration –**
- Collaboration is hallmark of cloud applications since they enable dispersed workgroups to come together on cloud platforms via shared usage and share information as well as collectively work on the data in real time.
- This facilitates the acceleration of product development and minimization of time required for marketing by improving customer service.

- **7) Cost-effectiveness –**
- Cloud service providers considerably mitigate organizations' expenditure on purchasing hardware components or equipment by taking care of these factors.
- The remotely positioned entire platform turns out to be more affordable than onsite data centers.

- **8) Instantaneous reduction of expenditure –**
- Organizations are able to face economic slowdown or recession, thanks to flexibility of cost structure.
- Cloud computing leverages pay-as-you-use model with no contractual obligations, so clients are not bound to pay any fee except for the resources deployed.

- **9) Optimization of capacity –**
- Users of cloud services are empowered with the flexibility of capacity that can be easily tuned up or down to enhance optimization.
- The examples can be seen in cloud hosting, dedicated server hosting, and many other related services.

- **1O) Eco-friendly –**
- Cloud computing is contributing to the global environment by reducing the number of data centers and improving work efficiency.
- As part of the global green initiative, cloud providers are mostly green hosts and make use of renewable energy sources as much as the energy-requirements allow.

COMPARISON OF GRID & UTILITY COMPUTING

- **1. Grid Computing:**
- Grid computing is also called as "**distributed computing**." It links multiple computing resources (PC's, workstations, servers, and storage elements) together and provides a mechanism to access them.
- The main advantages of grid computing is that it increases user productivity by providing transparent access to resources, and work can be completed more quickly.

- **2. Utility Computing :**

  Utility Computing, as name suggests, is a type of computing that provide services and computing resources to customers.

  It is basically a facility that is being provided to users on their demand and charge them for specific usage.

  It is similar to cloud computing and therefore requires cloud-like infrastructure.

Grid Computing	Utility Computing
It is a process architecture that combines different computing resources from multiple locations to achieve desired and common goal.	It is process architecture that provide on-demand computing resources and infrastructure on basis of pay per use method.
It distributes workload across multiple systems and allow computers to contribute their individual resources to common goal.	It allows organization to allocate and segregate computing resources and infrastructure to various users on basis of their requirements.
It makes better use of existing resources, address rapid fluctuations in customer demands, improve computational capabilities, provide flexibility, etc.	It simply reduces IT costs, easier to manage, provide greater flexibility, compatibility, provide more convenience, etc.

It mainly focuses on sharing computing resources.	It mainly focuses on acquiring computing resources.
It is of three types i.e., computational grid, data grid, and collaborative grid.	It is of two type i.e., Internal and external utility.
It is used in ATMs, back-end infrastructures, marketing research, etc.	It is used in large organizations such as Amazon, Google, etc., where they establish their own utility services for computing storage and applications.
Its main purpose is to integrate usage of computer resources from cooperating partners in form of VO (Virtual Organizations).	Its main purpose is to make computing resources and infrastructure management available to customer as per their need, and charge them for specific usage rather than flat rate.

Cloud computing and other computing systems

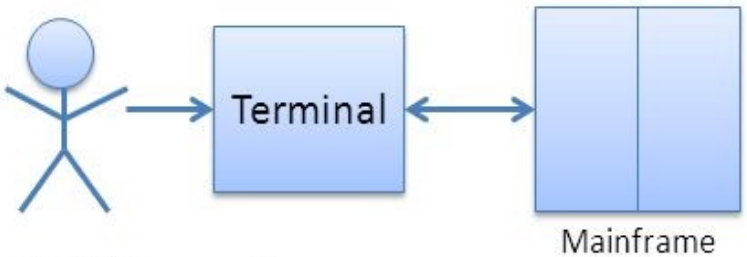
- Computing environments refer to the technology, infrastructure and software platforms that are used to develop, test, deploy, and run software applications.
- There are several types of computing environments, including:
- **Mainframe**: A large and powerful computer system used for critical applications and large-scale data processing.

- **Client-Server**: A computing environment in which client devices access resources and services from a central server.
- **Cloud Computing**: A computing environment in which resources and services are provided over the Internet and accessed through a web browser or client software.
- **Mobile Computing**: A computing environment in which users access information and applications using handheld devices such as smart phones and tablets.

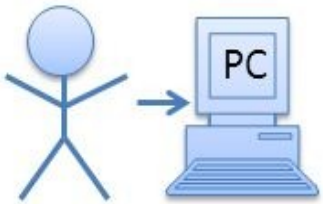
- **Grid Computing**: A computing environment in which resources and services are shared across multiple computers to perform large-scale computations.
- **Embedded Systems**: A computing environment in which software is integrated into devices and products, often with limited processing power and memory.

Computing Generations

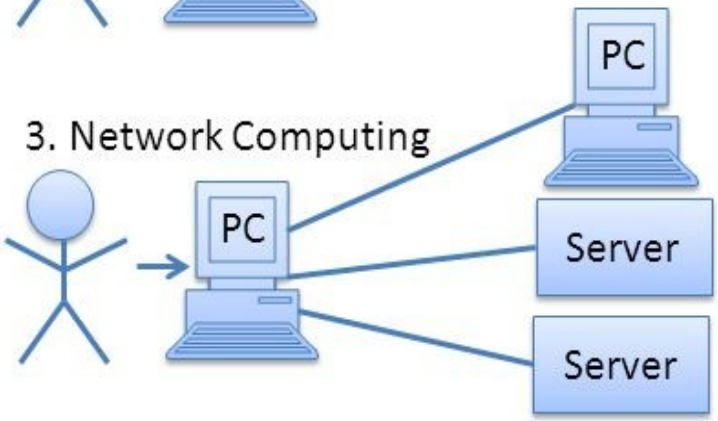
1. Mainframe Computing



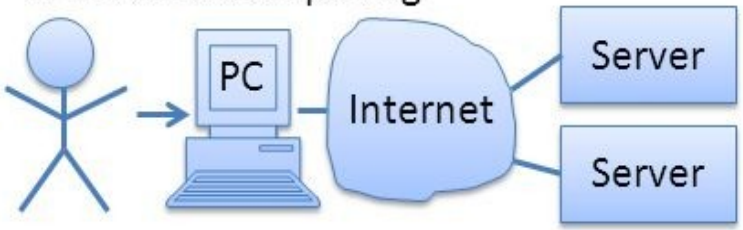
2. PC Computing



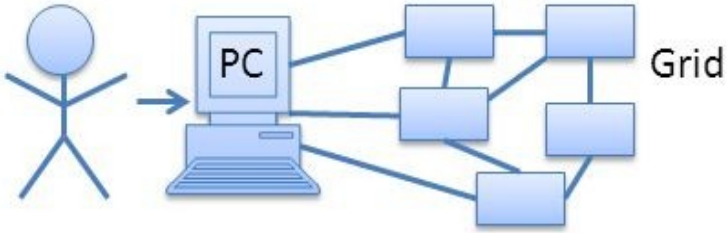
3. Network Computing



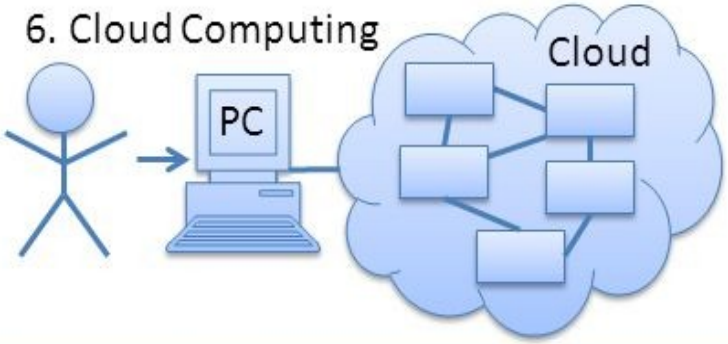
4. Internet Computing



5. Grid Computing



6. Cloud Computing



Advantages of different computing environments:

- **Mainframe:** High reliability, security, and scalability, making it suitable for mission-critical applications.
- **Client-Server**: Easy to deploy, manage and maintain, and provides a centralized point of control.
- **Cloud Computing**: Cost-effective and scalable, with easy access to a wide range of resources and services.

- **Mobile Computing**: Allows users to access information and applications from anywhere, at any time.
- **Grid Computing:** Provides a way to harness the power of multiple computers for large-scale computations.
- **Embedded Systems:** Enable the integration of software into devices and products, making them smarter and more functional.

Disadvantages of different computing environments:

- Mainframe: High cost and complexity, with a significant learning curve for developers.
- Client-Server: Dependence on network connectivity, and potential security risks from centralized data storage.
- Cloud Computing: Dependence on network connectivity, and potential security and privacy concerns.

- Mobile Computing: Limited processing power and memory compared to other computing environments, and potential security risks.
- Grid Computing: Complexity in setting up and managing the grid infrastructure.
- Embedded Systems: Limited processing power and memory, and the need for specialized skills for software development

APPLICATIONS OF CLOUD COMPUTING

- Cloud service providers provide various applications in the field of art, business, data storage and backup services, education, entertainment, management, social networking, etc.
- The most widely used cloud computing applications are given below -

Cloud Computing Application

- Art
- Business
- Data Storage and Backup
- Education
- Entertainment
- Management
- Social

- **1. Art Applications**
- Cloud computing offers various art applications for quickly and easily design **attractive cards, booklets,** and **images**. Some most commonly used cloud art applications are given below:
- **i Moo**
- Moo is one of the best cloud art applications. It is used for designing and printing business cards, postcards, and mini cards.

- **ii. Vistaprint**
- Vistaprint allows us to easily design various printed marketing products such as business cards, Postcards, Booklets, and wedding invitations cards.
- **iii. Adobe Creative Cloud**
- Adobe creative cloud is made for designers, artists, filmmakers, and other creative professionals. It is a suite of apps which includes PhotoShop image editing programming, Illustrator, InDesign, TypeKit, Dreamweaver, XD, and Audition.

- **2. Business Applications**
- Business applications are based on cloud service providers.
- Today, every organization requires the cloud business application to grow their business.
- It also ensures that business applications are 24*7 available to users.

- **i. MailChimp**
- MailChimp is an **email publishing platform** which provides various options to **design, send,** and **save** templates for emails.
- **ii. Salesforce**
- Salesforce platform provides tools for sales, service, marketing, e-commerce, and more. It also provides a cloud development platform.

- **iii PAYPAL**
- Paypal offers the simplest and easiest **online payment** mode using a secure internet account.
- Paypal accepts the payment through debit cards, credit cards, and also from Paypal account holders.

- **3. Data Storage and Backup Applications**
- Cloud computing allows us to store information (data, files, images, audios, and videos) on the cloud and access this information using an internet connection.
- As the cloud provider is responsible for providing security, so they offer various backup recovery application for retrieving the lost data.

- **i. Box.com**
- Box provides an online environment for **secure content management, workflow,** and **collaboration**. It allows us to store different files such as Excel, Word, PDF, and images on the cloud.
- The main advantage of using box is that it provides drag & drop service for files and easily integrates with Office 365, G Suite, Salesforce, and more than 1400 tools.

- **ii. Mozy**
- Mozy provides powerful **online backup solutions** for our personal and business data. It schedules automatically back up for each day at a specific time.
- **iii. Google G Suite**
- Google G Suite is one of the best **cloud storage** and **backup** application. It includes Google Calendar, Docs, Forms, Google+, Hangouts, as well as cloud storage and tools for managing cloud apps. The most popular app in the Google G Suite is Gmail. Gmail offers free email services to users.

- **4. Education Applications**
- Cloud computing in the education sector becomes very popular.
- It offers various **online distance learning platforms** and **student information portals** to the students.
- The advantage of using cloud in the field of education is that it offers strong virtual classroom environments,
- Ease of accessibility, secure data storage, scalability, greater reach for the students, and minimal hardware requirements for the applications.

- **i. Google Apps for Education**
- Google Apps for Education is the most widely used platform for free web-based email, calendar, documents, and collaborative study.
- **ii. Chromebooks for Education**
- Chromebook for Education is one of the most important Google's projects. It is designed for the purpose that it enhances education innovation.

- **5. Entertainment Applications**
- Entertainment industries use a **multi-cloud strategy** to interact with the target audience.
- Cloud computing offers various entertainment applications such as online games and video conferencing.

- **i. Online games**
- Today, cloud gaming becomes one of the most important entertainment media.
- It offers various online games that run remotely from the cloud.
- The best cloud gaming services are Shaow, GeForce Now, Vortex, Project xCloud, and PlayStation Now.

- **6. Management Applications**
- Cloud computing offers various cloud management tools which help admins to manage all types of cloud activities, such as resource deployment, data integration, and disaster recovery.
- These management tools also provide administrative control over the platforms, applications, and infrastructure.

- **i. Toggl**
- Toggl helps users to track allocated time period for a particular project.
- **ii. Evernote**
- Evernote allows you to sync and save your recorded notes, typed notes, and other notes in one convenient place. It is available for both free as well as a paid version.
- It uses platforms like Windows, macOS, Android, iOS, Browser, and Unix.

- **7. Social Applications**
- Social cloud applications allow a large number of users to connect with each other using social networking applications such as **Facebook, Twitter, LinkedIn,** etc.

- **i. Facebook**
- Facebook is a **social networking website** which allows active users to share files, photos, videos, status, more to their friends, relatives, and business partners using the cloud storage system.
- **ii. Twitter**
- Twitter is a **social networking** site. It is a **microblogging** system. It allows users to follow high profile celebrities, friends, relatives, and receive news. It sends and receives short posts called tweets.

IT as a service

- IT as a service, or ITaaS, is a cloud-based technology service that allows businesses to access high-quality software and hardware without having to purchase, maintain, or upgrade it.
- Instead of managing all their own equipment and software, companies can leverage a provider’s infrastructure to give access to services such as network security, storage, and backup.

What are IT services

- IT services are a range of services that help customers manage their technology.
- They include everything from software maintenance and support to data backup and recovery.
- Here are some examples of IT services:
- Network design and implementation
- Endpoint security
- Cloud services
- Data center operations and management
- Security administration

Benefits of ITaaS

- Cost savings: ITaaS can reduce operational costs, hardware costs, and maintenance costs.
- Flexibility: ITaaS can scale up or down as a business's needs change.
- Focus on core business: ITaaS allows businesses to focus on innovation and responding to market demands instead of infrastructure investments.
- Standardized processes: ITaaS can streamline operations with standardized processes.

- Cybersecurity: ITaaS can protect all aspects of a business's technology systems.
- 24/7 support: ITaaS can provide help desk services around the clock.
- Access to expertise: ITaaS can provide access to expertise from IT service providers.
- Improved compliance: ITaaS can help businesses improve compliance and security.

Workload Patterns for cloud computing

- **What Is a Cloud Workload?**
- A cloud workload refers to the computing resources and tasks that are required to run an application or service in a cloud computing environment.
- This can include resources such as virtual machines, storage, and networking, as well as the software and applications that run on those resources.

Following are the workload patterns for cloud computing

- On and Off
- Bursting Predictable and Unpredictable
- Growing Fast or Scale cheap, Fail cheap
- Connected applications
- Commodity workloads.

- **On and Off**

- Applications or workloads that have a relatively short period of activity and after they have produced a result can be switched of.
- Batch processing, data crunching or calculations are typical activities of applications of this pattern.
- The obvious advantage for workloads with this pattern is that **you don’t have to pay for hardware when it is sitting around idle.**

- **Bursting Predictable and Unpredictable**
- When you are running a successful web retail operation you probably have your peak demand during the holiday season.
- After a couple of years you will have a fairly good idea how much bigger the demand will be during this period than during the rest of the year.
- This can be up to 800% higher than any other day of the year.

- **Growing Fast or Scale cheap, Fail cheap**

- This is the scenario of the tech startup. Imagine a service that allows you in a very short message (say 140 characters) that allows people to know what they are doing or thinking.
- Suddenly your service becomes a great hit and the infrastructure you are running it one can’t cope.
- You are not able to scale your infrastructure to the demand and the service breaks when it is being most used.
- The problem is that you never know if your service will be such a success.

- **Connected applications**
- Another workload pattern I think has great benefits from running in the cloud are connected applications.
- In other words, most cloud platforms are located on the internet and are therefore easy to connect to from any other application or device that is connected to the internet.

- **Commodity workloads**
- The workloads described above are workloads of custom application that would be deployed to a platform as a service solution like Windows Azure, Amazon EC2 or Google App Engine.
- There are workloads that are so common,place that migrating them to a cloud platform makes sense. E-mail is such a workload.
- There is nothing strategical about running your own E-mail platform so move it to the cloud. Buy it as a Service.